

Exam

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Electricity and Electronics EBN 122

12 November 2019

Test Information

Maximum marks:	32	Full marks:	45
Duration of test:	90 min + (5 min before and 10 min after for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)		17	

Important

1. The examination regulations of the University of Pretoria apply.
2. Answer all questions on the OCR forms provided. The question number **01** in this question paper refers to row 01 of the OCR form.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.
6. Use at least 3 significant numbers to write irrational numbers.

Lecturers: Prof J.P. de Villiers, Prof J.P. Jacobs, Dr X. Ye**INSTRUCTIONS****Do step 0 in the first 5 minutes of the session. (-5 to 0 min)****Do step 1 in the next 90 minutes of the test. (0-90 min)****Do step 2 within the next 10 minutes. (90-100 min)**

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR form.

STEP 1 (0 min to 90 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

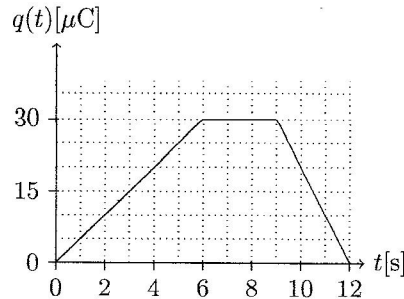
STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2019EBN122E01**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1 / Afdeling 1

Question 1 [1]

The graph of the charge in an uncharged capacitor is shown in the figure below.



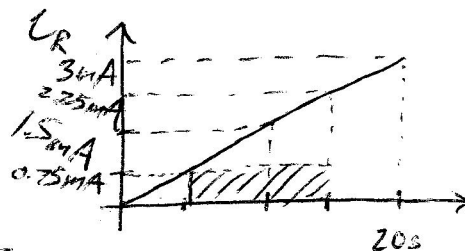
01 Find the current at $t = 10$ s. [1]

$$\begin{aligned}
 i_c &= \frac{dq_c(t)}{dt} \text{ at } t=10\text{s} \\
 &= \frac{\Delta y}{\Delta x} = -\frac{30 \times 10^{-6}}{3} = -10 \times 10^{-6} \text{ A} \rightarrow
 \end{aligned}$$

Question 2 [2]

The voltage across a 5Ω resistor increases linearly from 0 mV to 15 mV between times $t = 0$ s and $t = 20$ s.

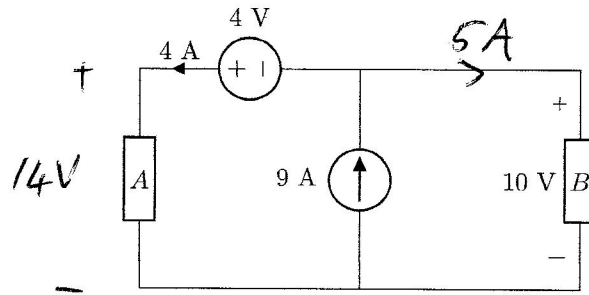
02 Calculate the charge that flows through the resistor between $t = 5$ s and $t = 15$ s. [2]



$$\begin{aligned}
 Q_R &= \int_5^{15} i_R(t) dt = 0.75 \times 10 + 5 \times (1.5) \text{ [mC]} \\
 &= 7.5 + 7.5 \text{ [mC]} \\
 &= 15 \text{ mC} \rightarrow
 \end{aligned}$$

Questions 3 to 5 [3]

Calculate the power relating to the following elements in the circuit below:

03 Calculate the power P_A associated with element A. [1]

04 Is element A supplying power (write a 1) or absorbing power (write a 2)? [1]

05 Calculate the resistance R_B of circuit element B. [1]03

$$V_A = 10 + 4 = 14 \text{ V}$$

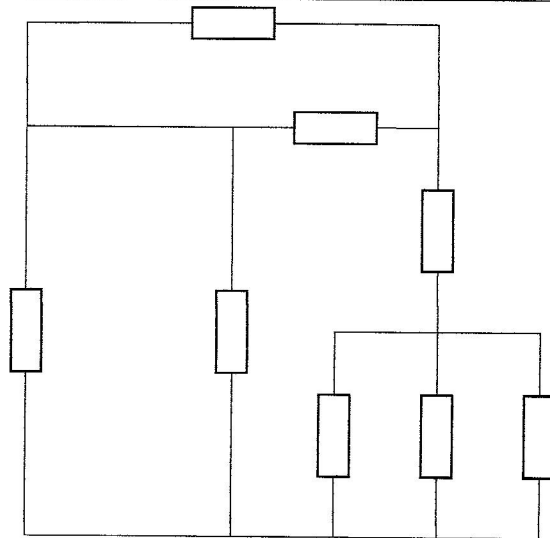
$$I_A = 4 \text{ A}$$

$$P_A = V_A I_A = 56 \text{ W} \rightarrow$$

04Absorbing power - 2 $\rightarrow$ 05

$$R_B = \frac{V_B}{I_B} = \frac{10}{5} = 2 \Omega \rightarrow$$

Questions 6 to 8 [3]



06	Determine the number of nodes n in the circuit above. [1]
07	Determine the number of branches b in the circuit above. [1]
08	Determine the number of independent loops l in the circuit above. [1]

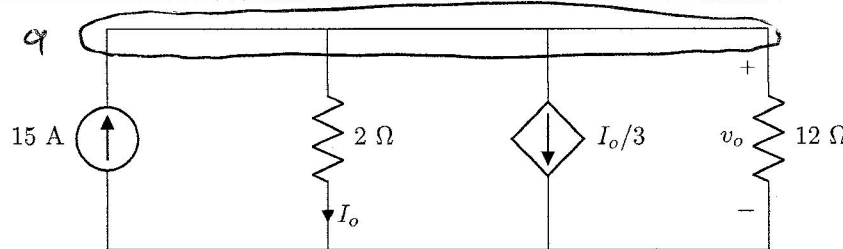
$$b = l + n - 1$$

Q6 $n = 4$

Q7 $b = 8$

Q8 $l = 5$

Test $8 = 5 + 4 - 1$ ✓

Questions 9 to 10 [3]	
	
09	Determine v_o . [2]
10	Determine i_o [1]

$$\text{KCL at } a: \quad I_o + I_o/3 + \frac{V_o}{12} = 15 \quad (1)$$

$$\text{but } I_o = \frac{V_o}{2} \quad (2)$$

$$(2) \text{ in } (1): \quad \frac{V_o}{2} + \frac{V_o}{6} + \frac{V_o}{12} = 15$$

$$(\times 12) \quad 6V_o + 2V_o + V_o = 15 \times 12$$

$$9V_o = 180$$

$$V_o = 20 \text{ V} \rightarrow$$

$$I_o = \frac{V_o}{2}$$

$$= 10 \text{ A} \rightarrow$$

$\Rightarrow$ Will mark with error.

Questions 11 to 12 [3]	
11	If $v_1 = 3$ V, determine R . [2]
12	If $v_x = 1.5$ V, determine I . [1]

11 Voltage divider:

$$v_2 = \frac{R}{R + 2 \parallel 2} 4 = 3$$

$$\frac{R}{R + 1} = \frac{3}{4}$$

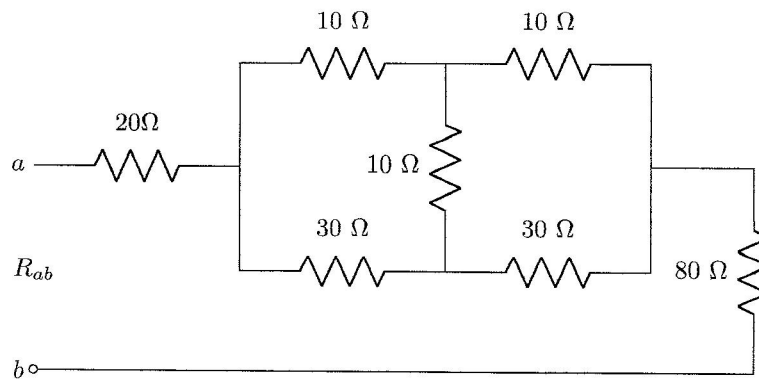
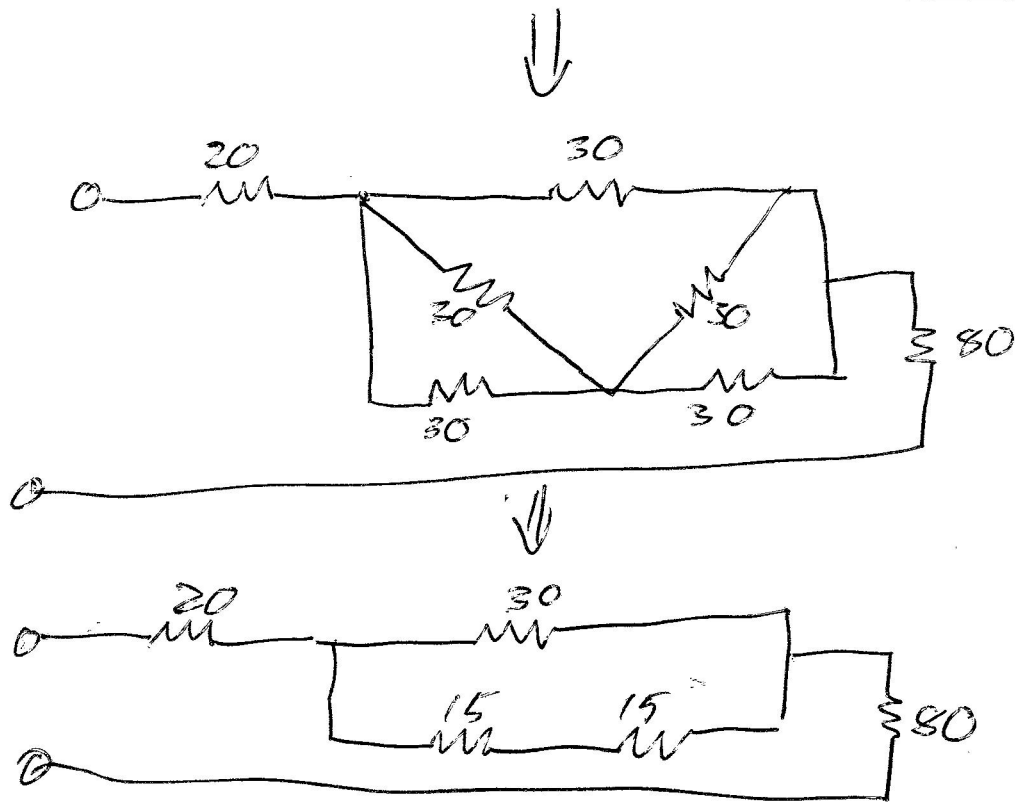
$$4R = 3R + 3 \quad \therefore R = 3 \Omega \rightarrow$$

12 Current divider:

$$9 \parallel 18 = \frac{9 \times 18}{9 + 18} = 6 \Omega$$

$$I = \frac{6}{6 + 3} \times 3(1.5) = \frac{6}{9} \times 4.5 = 3 A \rightarrow$$

Question 13 [2]

13 Determine R_{ab} . [2]

$$R_{ab} = 20 + 30 // 30 + 80 = 20 + 15 + 80 = 115 \Omega$$

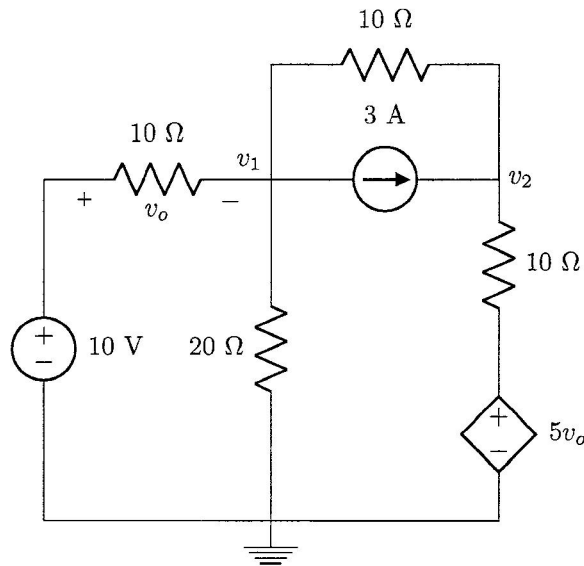
Questions 14 to 17 [4]

Use nodal analysis to set up the following two equations of the form

$$av_1 - 2v_2 = b,$$

$$cv_1 + dv_2 = -50.$$

Calculate the unknown ~~integer~~ coefficients.



14 a [1]

15 b [1]

16 c [1]

17 d [1]

Node v_1 : $\frac{v_1 - 10}{10} + \frac{v_1 - v_2}{10} + \frac{v_1}{20} + 3 = 0$

($\times 20$) $2v_1 - 20 + 2v_1 - 2v_2 + v_1 + 60 = 0$

Alternative $5v_1 - 2v_2 = -40 \rightarrow$
 $(6.25v_1 - 2.5v_2 = -50)$

Node v_2 : $\frac{v_1 - v_2}{10} + 3 = \frac{v_2 - 5v_o}{10}$

($\times 10$) $v_1 - v_2 + 30 = v_2 - 5v_o$, but $v_o = 10 - v_1$ (2)

(2) in (1) $v_1 - v_2 + 30 = v_2 - 5(10 - v_1)$

$v_1 - v_2 + 30 = v_2 - 50 + 5v_1$

Alternative $(-4v_1 - 2v_2 = -80)$

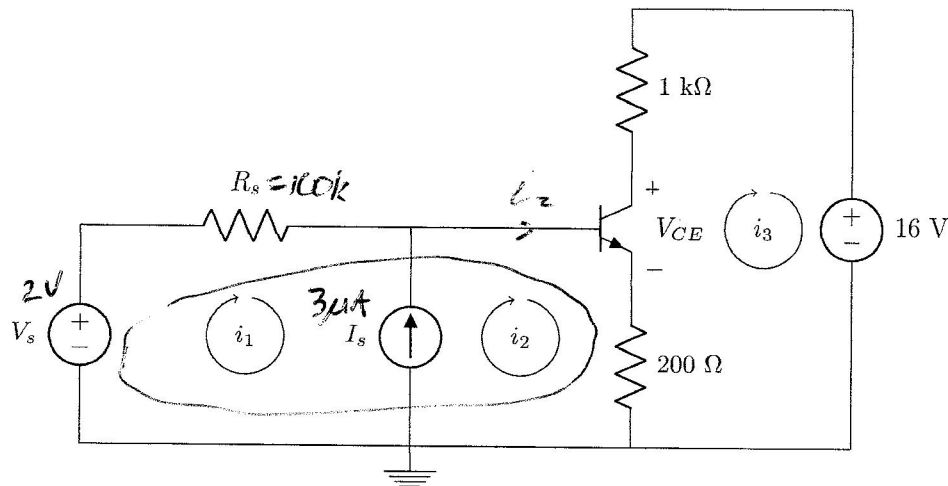
($\times \frac{5}{8}$) $-2.5v_1 - 1.25v_2 = -50$

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$a = 5, b = -40, c = -2.5, d = -1.25$ or

Questions 18 to 22 [5]

Consider the circuit below containing a transistor with $\beta = 99$ and $V_{BE} = 0.7$ V.



Assuming $V_s = 2$ V, $R_s = 100$ k Ω and $I_s = 3$ μ A, use mesh analysis to set up the following two equations of the form

$$ei_1 + fi_2 = 1.3,$$

$$gi_1 + hi_2 = 3.0 \times 10^{-6}.$$

Calculate the unknown coefficients e , f , g and h .

18	e [1]	19	f [1]	20	g [1]	21	h [1]
22	Determine the value of V_{CE} if $i_2 = 13.333$ μ A. [1]						

KVL supermesh: $-2 + 100 \times 10^3 i_1 + 0.7 + 200(i_2 - i_3) = 0$

$$\therefore -1.3 + 100 \times 10^3 i_1 + 200(i_2 - i_3) = 0 \quad (1)$$

But $i_3 = -(\beta + 1)i_2 \therefore i_3 = -100i_2$ (2)

(2) in (1) $100 \times 10^3 i_1 + 200 i_2 - 200(-100 i_2) = 1.3$

$$\therefore 100 \times 10^3 i_1 + 2.02 \times 10^4 i_2 = 1.3$$

Supermesh

$$i_2 - i_1 = 3 \times 10^{-6}$$

$$-i_1 + i_2 = 3 \times 10^{-6}$$

$$e = 100 \times 10^3$$

$$f = 2.02 \times 10^4$$

$$g = -1$$

$$h = 1$$

22 KVL loop 3: $V_{ik} = \beta i_2 (1000) = 1.32$ V

$$V_{200} = (\beta + 1)i_2 (200) = 0.267$$
 V

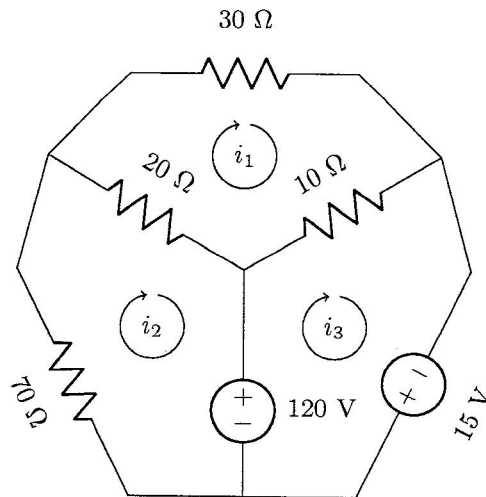
$$V_{CE} = 16 - 1.32 - 0.267 = 14.41$$
 V

Questions 23 to 26 [4]

Use inspection to compile a matrix equation of the form below

$$\begin{bmatrix} i & j & l \\ j & m & 0 \\ l & 0 & p \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -120 \\ r \end{bmatrix}$$

and provide the values of the matrix elements asked for below.



23	i [1]	24	m [1]	25	l [1]	26	r [1]
----	---------	----	---------	----	---------	----	---------

$$i = 60$$

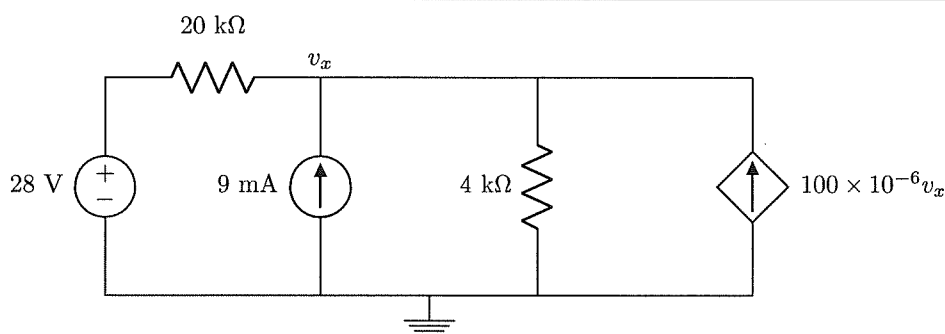
$$m = 90$$

$$l = -10$$

$$r = 135$$

Section 2 / Afdeling 2

Questions 27 to 28 [4]



27 Determine the contribution to v_x of the 28 V voltage source. [2]

28 Determine the contribution to v_x of the 9 mA current source. [2]

(27)

Deactivate current source:

$$\frac{v_x' - 28}{20000} + \frac{v_x'}{4000} - 100 \times 10^{-6} v_x' = 0$$

$$\frac{v_x' - 28}{2} + \frac{v_x'}{0.4} - v_x' = 0$$

$$v_x' - 56 + 5v_x' - 2v_x' = 0 \quad \therefore \underline{v_x' = 14 \text{ V}}$$

(28)

Deactivate voltage source:

$$\frac{v_x''}{20000} - 9 \times 10^{-3} + \frac{v_x''}{4000} - 100 \times 10^{-6} v_x'' = 0$$

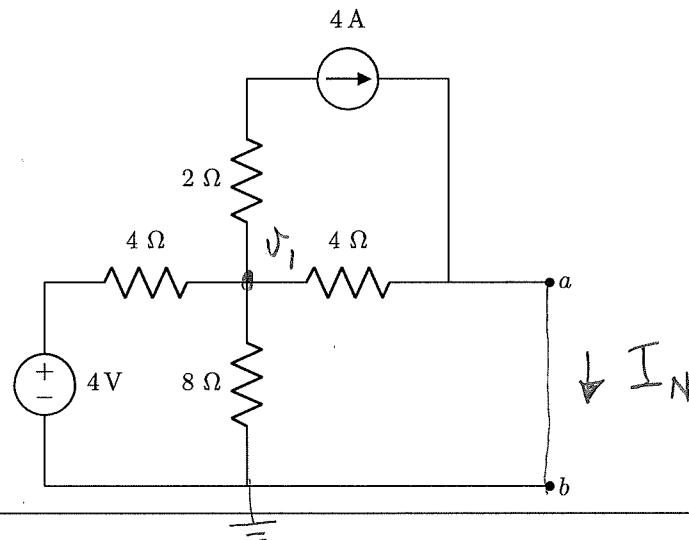
$$\frac{v_x''}{2} - 90 + \frac{v_x''}{0.4} - v_x'' = 0$$

$$v_x'' - 180 + 5v_x'' - 2v_x'' = 0$$

$$\therefore \underline{v_x'' = 45 \text{ V}}$$

Questions 29 to 30 [4]

Find the Norton equivalent current I_N , and Norton equivalent resistance R_N , with reference to terminals $a - b$.



29 I_N [2]

30 R_N [2]

$$\textcircled{29} \quad \frac{v_1 - 4}{4} + \frac{v_1}{8} + 4 + \frac{v_1}{4} = 0$$

$$2v_1 - 8 + v_1 + 32 + 2v_1 = 0$$

$$5v_1 = -24 \therefore v_1 = -4.8 \text{ V}$$

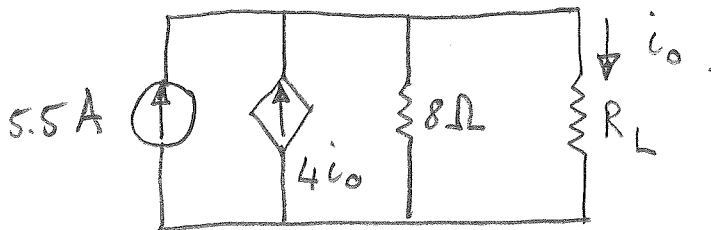
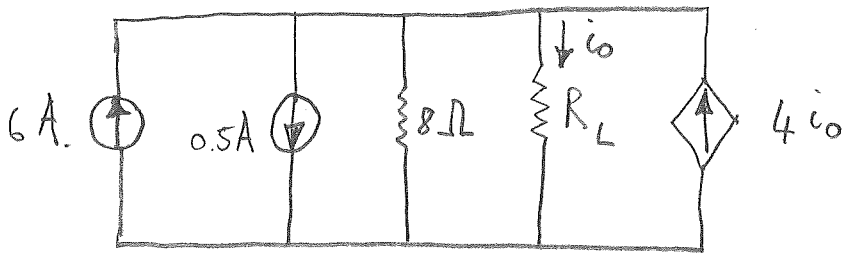
$$I_N = 4 + \frac{v_1}{4} = \underline{2.8 \text{ A}}$$

$$\textcircled{30} \quad R_N = 4 + 4 \parallel 8 = 6.667 \, \Omega$$

Questions 31 to 33 [4]

Simplify the circuit on the left by means of source transformations so that it looks like the circuit on the right. Determine R_x , A and B .

31	R_x [2]
32	A [1]
33	B [1]

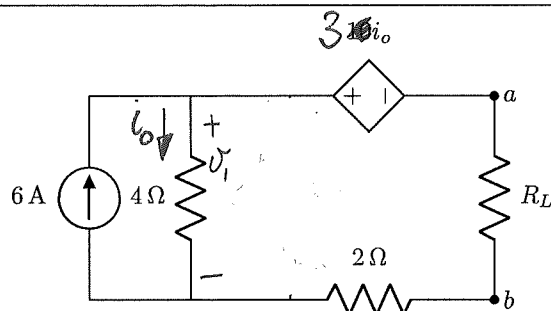


$$\begin{aligned}
 A &= 4 \\
 B &= 5.5 \\
 R_x &= 8\ \Omega
 \end{aligned}$$



Questions 34 to 35 [4]

Find the Thevenin equivalent voltage V_{Th} seen by the load R_L , and the value of R_L so that maximum power is transferred to R_L .



34 V_{Th} [2]

35 R_L [2]

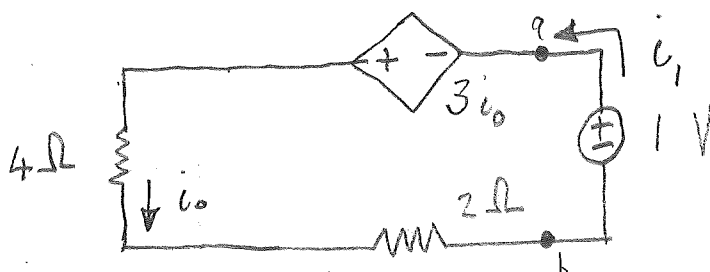
34

$$v_1 = 6 \cdot 4 = 24 \text{ V}$$

$$i_0 = 6 \text{ A}$$

$$V_{Th} = v_1 - 3i_0 = 24 - 18 = 6 \text{ V} \rightarrow$$

35

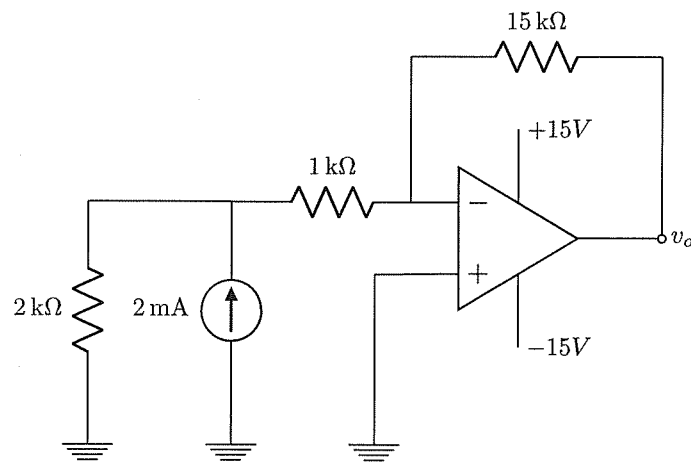
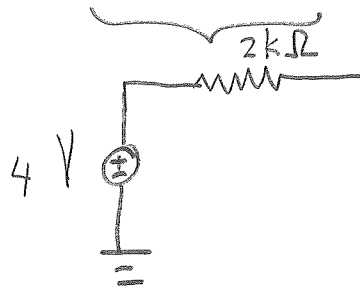


$$\text{KVL: } -1 - 3i_0 + 4i_0 + 2i_0 = 0$$

$$3i_0 = 1 \therefore i_0 = 0.333 \text{ A} = i_1$$

$$\therefore R_L = R_{Th} = \frac{1}{i_1} = 3 \Omega \rightarrow$$

Question 36 [2]

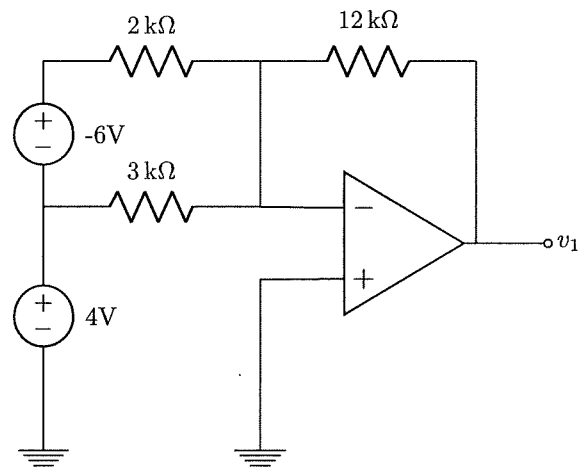
Assume that the op-amp is ideal. Find v_o .36 v_o [2]

$$v_o = - \frac{15000}{2000 + 1000} 4$$

$$= -20 \text{ V}$$

But op-amp saturates $\therefore v_o = -15 \text{ V}$

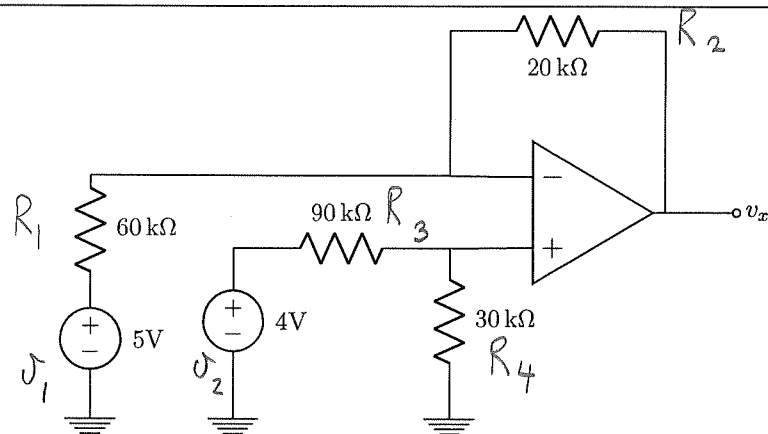
Question 37 [2]

Assume that the op-amp is ideal. Find v_1 .37 v_1 [2]

Summing amplifier:

$$v_1 = - \left(\frac{12000}{2000} (-6 + 4) + \frac{12000}{3000} (4) \right)$$
$$= -4\text{V}$$

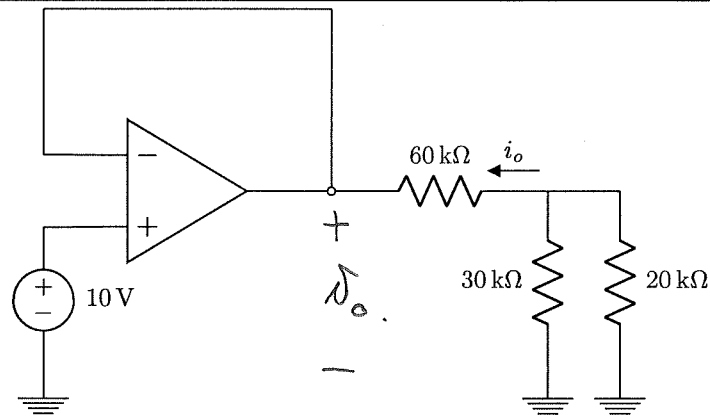
Question 38 [2]

Assume that the op-amp is ideal. Find v_x .38 v_x [2]

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \Rightarrow \text{difference amplifier}$$

$$\begin{aligned} \therefore v_x &= \frac{R_2}{R_1} (v_2 - v_1) \\ &= \frac{20}{60} (4 - 5) \\ &= -\frac{1}{3} \text{ V} \end{aligned}$$

Question 39 [2]

Assume that the op-amp is ideal. Find i_o .39 i_o [2]

Voltage follower $\therefore v_o = 10 \text{ V}$

$$i_o = - \frac{v_o}{\left(60 + \frac{30 \cdot 20}{30 + 20}\right) \times 10^3}$$

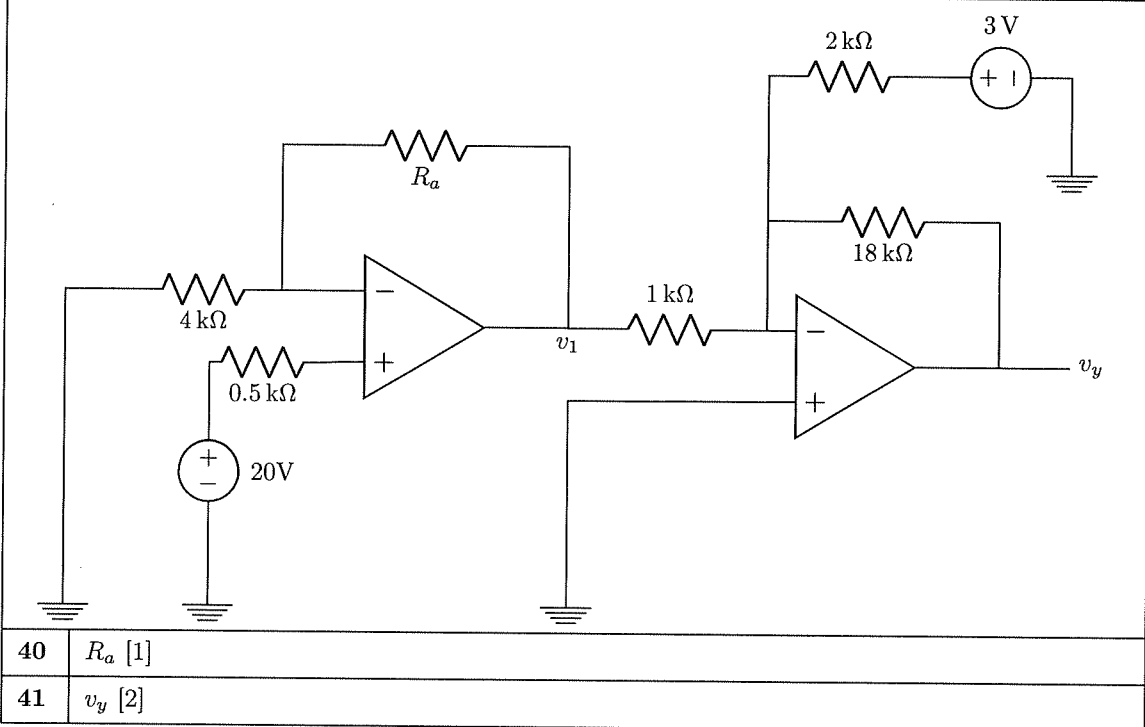
$$= - \frac{10}{(60 + 12) \times 10^3}$$

$$= 0.1389 \text{ mA}$$



Questions 40 to 41 [3]

Two ideal op-amps are connected as shown. It is known that $v_1 = 30$ V. Find R_a and v_y .



④① Left circuit: non-inverting op-amp

$$v_1 = \left(1 + \frac{R_a}{4000}\right) 20$$

$$30 = 20 + 20 \frac{R_a}{4000}$$

$$10 = 20 \frac{R_a}{4000} \quad \therefore R_a = 2 \text{ k}\Omega$$

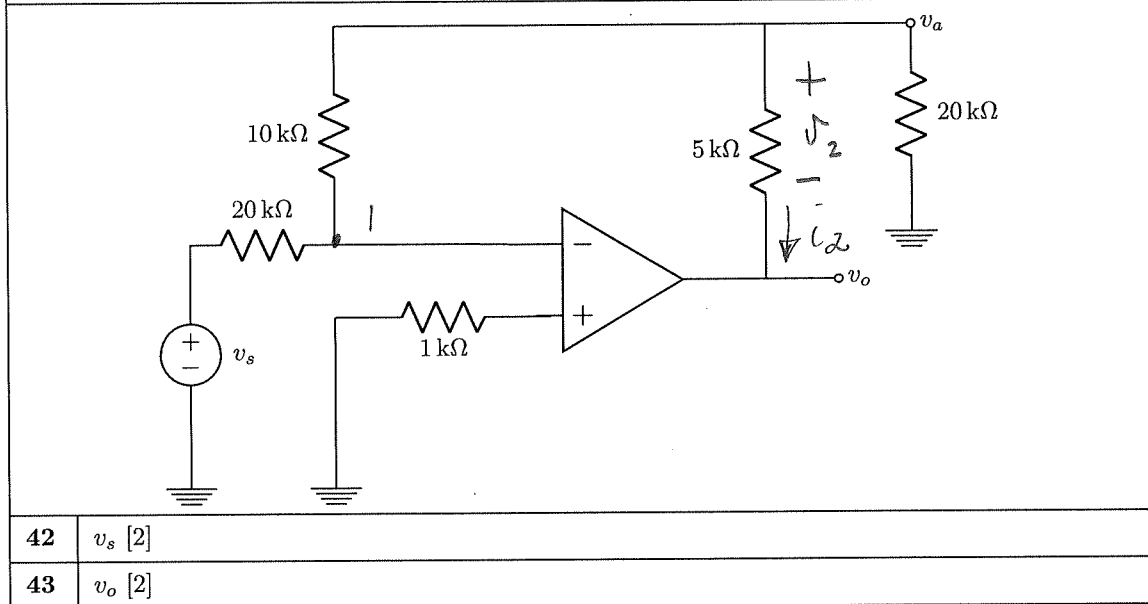
④② Right circuit: summing amplifier

$$v_y = - \left(\frac{18}{2} 3 + \frac{18}{1} v_1 \right)$$

$$= - \left(\frac{18}{2} 3 + \frac{18}{1} 30 \right)$$

$$= -567 \text{ V}$$

Questions 42 to 43 [4]

Assume that the op-amp is ideal, and that $v_a = 10$ V. Find v_s and v_o .42 v_s [2]43 v_o [2]

42) KCL at 1:

$$\frac{0 - v_s}{20} + \frac{0 - v_a}{10} = 0$$

$$-v_s - 2(10) = 0 \quad \therefore \underline{v_s = -20 \text{ V}}$$

43)

$$i_2 + \frac{v_a}{20000} = \frac{0 - v_a}{10000}$$

$$\therefore i_2 = \frac{0 - 10}{10000} - \frac{10}{20000} = -1.5 \text{ mA}$$

$$v_2 = 5000 \cdot i_2 = -7.5 \text{ V}$$

$$v_o + v_2 = v_a$$

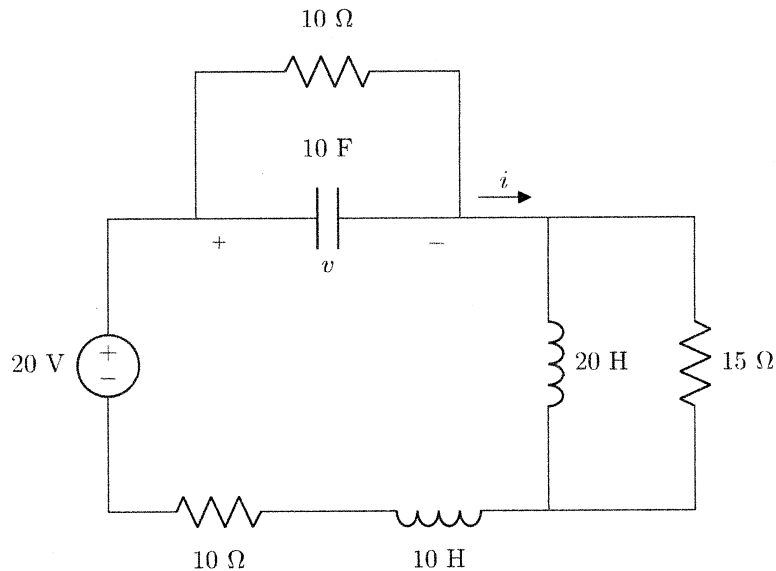
$$\therefore \underline{v_o = 17.5 \text{ V}}$$

Total Section 2 / Totaal Afdeling 2: [31]

Section 3

Questions 44 to 47 [4]

The circuit below operates under steady-state dc conditions.

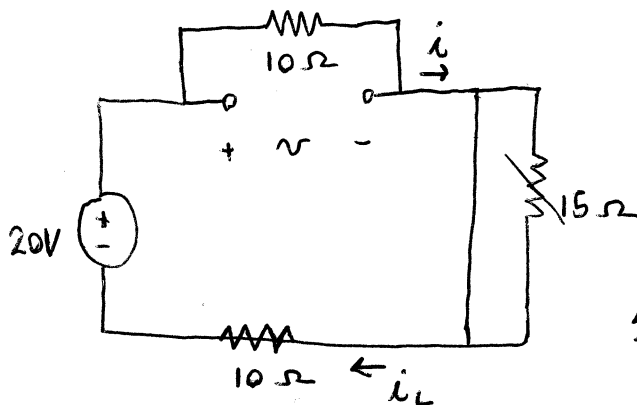


44 Determine i . [1]

45 Determine v . [1]

46 Determine the energy E_L stored in the 10 H inductor. [1]

47 Determine the energy E_C stored in the capacitor. [1]



$$\bullet \quad \underline{44}: i = \frac{20V}{10\Omega + 10\Omega} = 1A \rightarrow$$

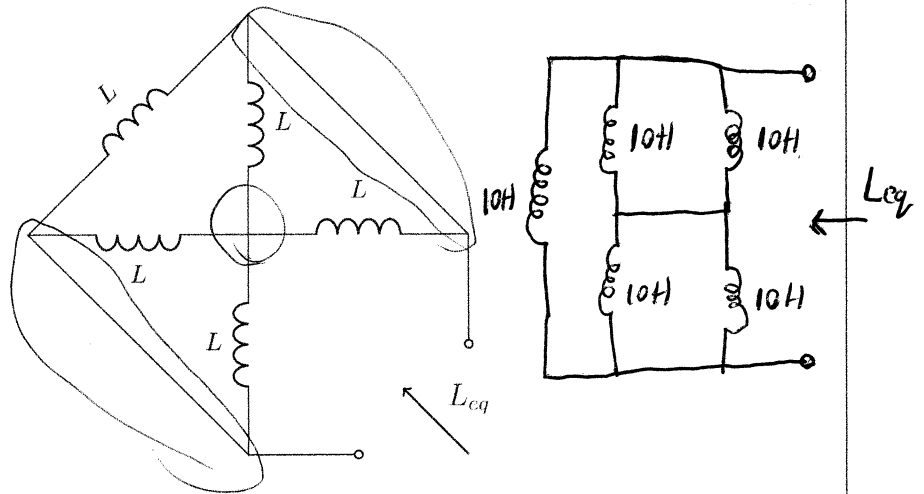
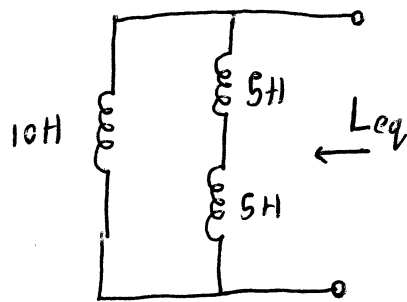
$$\underline{45}: v = iR = (1A)(10\Omega) = 10V \rightarrow$$

$$\begin{aligned} \underline{46}: E_L &= \frac{1}{2} L i^2 \\ &= \frac{1}{2} (10) (1)^2 \\ &= 5J \rightarrow \end{aligned}$$

$$\begin{aligned} \underline{47}: E_C &= \frac{1}{2} C v^2 \\ &= \frac{1}{2} (10) (10)^2 \\ &= 500J \rightarrow \end{aligned}$$

Question 48 [2]

Consider the circuit below.

48 Find L_{eq} if $L = 10$ H. [2]

$$L_{eq} = \left(\frac{1}{10} + \frac{1}{5+5} \right)^{-1}$$

$$= 5 \text{ H}$$

Questions 49 to 51 [4]	
Consider three sinusoid waveforms $v_1(t)$, $v_2(t)$, and $v_3(t)$, where $v_1(t) = 5 \sin(\omega t + 30^\circ), \text{ and}$ $v_2(t) = 12 \cos(\omega t + 60^\circ).$ The waveforms $v_1(t)$, $v_2(t)$, and $v_3(t)$ can be denoted by phasors $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$, respectively. The phase angle between $\mathbf{V}_2$ and $\mathbf{V}_3$ is ϕ_0, and the phase angle between $\mathbf{V}_1$ and $\mathbf{V}_3$ is $3\phi_0$; $\mathbf{V}_3$ is in the 1st quadrant of the phasor domain.	
49	Choose the correct statement. 1. v_2 leads v_3. 2. v_1 leads v_3. 3. v_1 leads v_2. Write 1 if statement 1 is correct, 2 if statement 2 is correct, or 3 if statement 3 is correct. [2]
50	Find $\phi_0 \in [0^\circ, 180^\circ]$. [1]
51	Use phasors to find $\mathbf{V} = \mathbf{V}_1 - \mathbf{V}_2$ in polar form. [1]

$$v_1(t) = 5 \sin(\omega t + 30^\circ) = 5 \cos(\omega t - 60^\circ)$$

$$\bar{V}_1 = 5 \angle -60^\circ$$

$$v_2(t) = 12 \cos(\omega t + 60^\circ)$$

$$\bar{V}_2 = 12 \angle 60^\circ$$

49: $\bar{V}_2$ leads $\bar{V}_3$

$$\phi_{V_2} - \phi_{V_3} = \phi_0$$

$$+ \quad -\phi_{V_1} + \phi_{V_3} = 3\phi_0$$

$$\phi_{V_2} - \phi_{V_1} = 4\phi_0$$

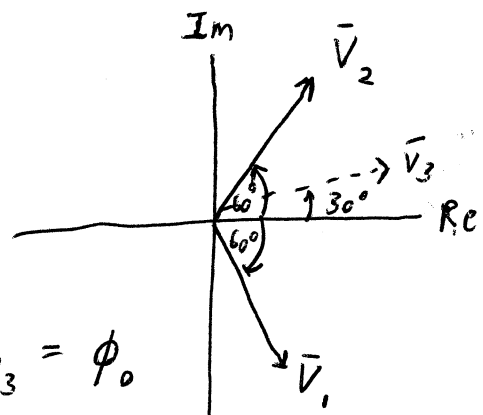
$$60 - (-60^\circ) = 4\phi_0$$

$$\underline{50}: \phi_0 = 30^\circ$$

51: $\bar{V} = \bar{V}_1 - \bar{V}_2$

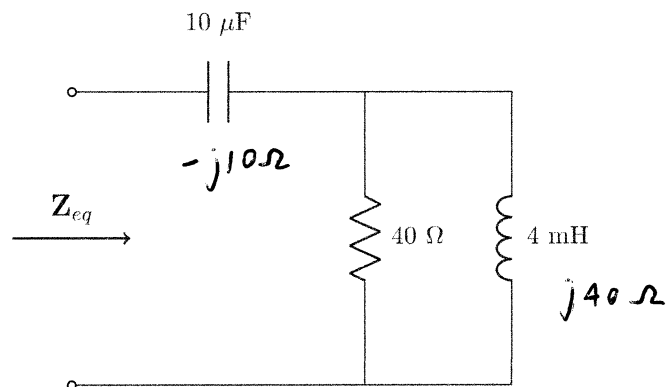
$$= 5 \angle -60^\circ - 12 \angle 60^\circ$$

$$\approx 10,44 \angle 35,50^\circ$$



Question 52 [2]

Consider the circuit below.

52 Find Z_{eq} in rectangular form at an angular frequency $\omega = 10^4\ \text{rad/s}$. [2]

$$X_C = \frac{1}{j\omega C} = \frac{1}{j(10000)(10 \times 10^{-6})} = -j10\ \Omega$$

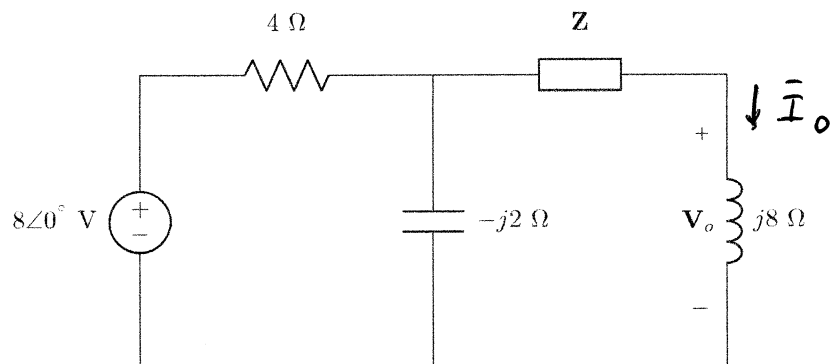
$$X_L = j\omega L = j(10000)(4 \times 10^{-3}) = j40\ \Omega$$

$$\bar{Z}_{eq} = -j10 + \left(\frac{1}{40} + \frac{1}{j40} \right)^{-1}$$

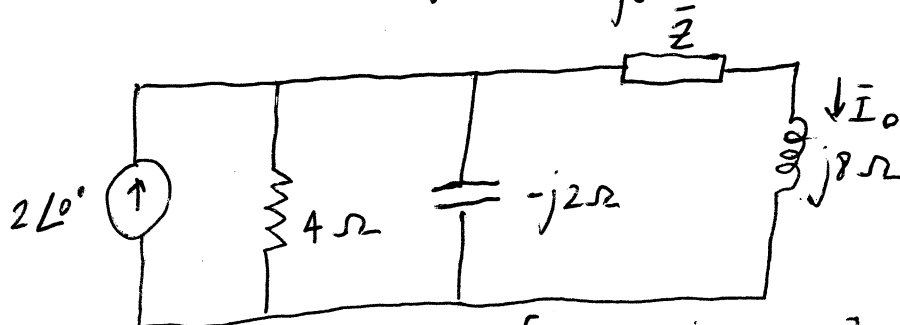
$$= 20 + j10\ \Omega$$

Question 53 [2]

Consider the circuit below.

53 Find $\mathbf{Z}$ when $\mathbf{V}_o = 16\angle 0^\circ$ in rectangular form. [2]

$$\bar{I}_o = \frac{\bar{V}_o}{j8} = \frac{16\angle 0^\circ}{j8} = -j2 \text{ A}$$



$$-j2 = 2 \left[\frac{0,8 - j1,6}{0,8 - j1,6 + \bar{Z} + j8} \right]$$

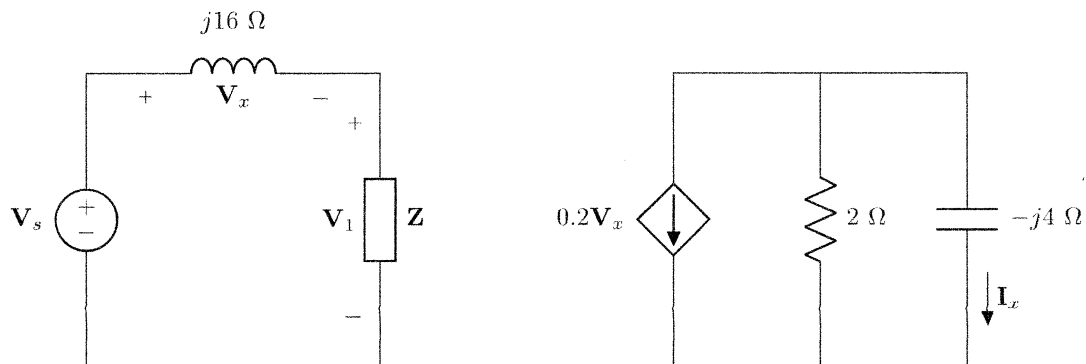
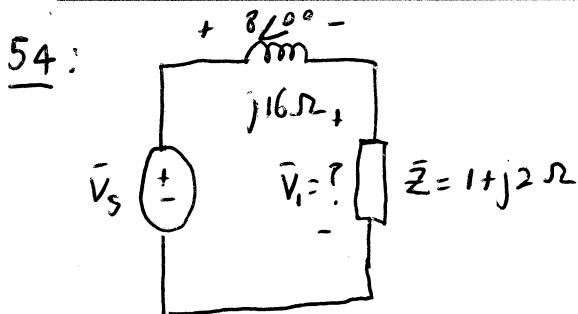
$$-j(0,8 - j1,6 + \bar{Z} + j8) = 0,8 - j1,6$$

$$\bar{Z} = 0,8 - j5,6 \Omega$$

→

Questions 54 to 55 [4]

Consider the circuit below.

54 Given $Z = 1 + j2 \Omega$ and $V_x = 8\angle 0^\circ$ V, find V_s in rectangular form. [2]55 If $I_x = 4\angle 0^\circ$ A, find V_x in rectangular form. [2]

$$\bar{I} = \frac{8\angle 0^\circ}{j16} = -\frac{j}{2} \text{ A}$$

$$-\bar{V}_s + 8 + 1 - j0,5 = 0$$

$$\bar{V}_s = 9 - j0,5 \text{ V}$$

$$\begin{aligned} \bar{V}_1 &= \bar{Z} \bar{I} \\ &= \left(-\frac{j}{2}\right)(1 + j2) \\ &= 1 - j0,5 \text{ V} \end{aligned}$$

55:

$$\bar{I}_x = -0,2 \bar{V}_x \left(\frac{2}{2 - j4} \right) = 4\angle 0^\circ$$

$$\bar{V}_x = -20 + j40 \text{ V}$$

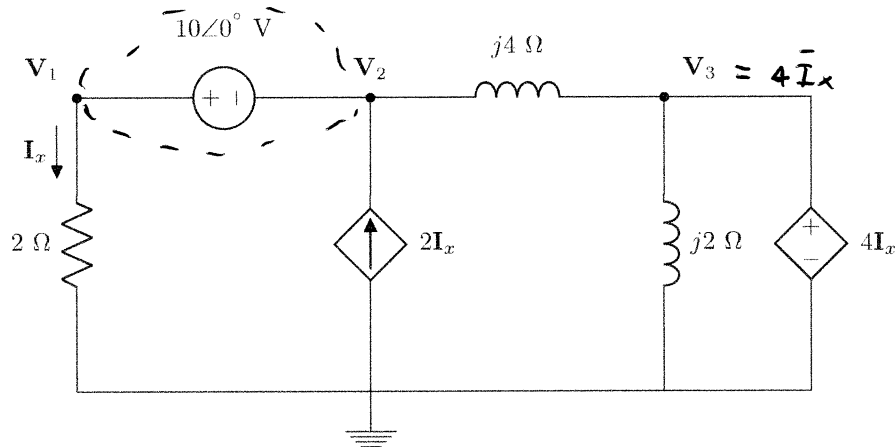
Questions 56 to 59 [4]

Use nodal analysis to set up two equations of the following form:

$$\mathbf{aV}_1 + \mathbf{bV}_2 = 10\angle 0^\circ,$$

$$\mathbf{cV}_1 + \mathbf{dV}_2 = \mathbf{d}.$$

Find the complex values $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, and $\mathbf{d}$ in rectangular form.



56	Determine $\mathbf{a}$. [1]
57	Determine $\mathbf{b}$. [1]
58	Determine $\mathbf{c}$. [1]
59	Determine $\mathbf{d}$. [1]

$$\bar{I}_x = \frac{\bar{V}_1}{2}$$

$$2\bar{I}_x = \bar{V}_1$$

$$4\bar{I}_x = 2\bar{V}_1$$

$$\bar{V}_2 + 10\angle 0^\circ = \bar{V}_1$$

$$\boxed{\bar{V}_1 - \bar{V}_2 = 10\angle 0^\circ}$$

$$\bar{a} = 1$$

$$\bar{b} = -1$$

$$\frac{\bar{V}_1}{2} - 2\bar{I}_x + \frac{\bar{V}_2 - 4\bar{I}_x}{j4} = 0$$

$$\frac{\bar{V}_1}{2} - \bar{V}_1 + \frac{\bar{V}_2 - 2\bar{V}_1}{j4} = 0$$

$$\bar{V}_1 \left[\frac{1}{2} - 1 - \frac{2}{j4} \right] + \bar{V}_2 \left[\frac{1}{j4} \right] = 0$$

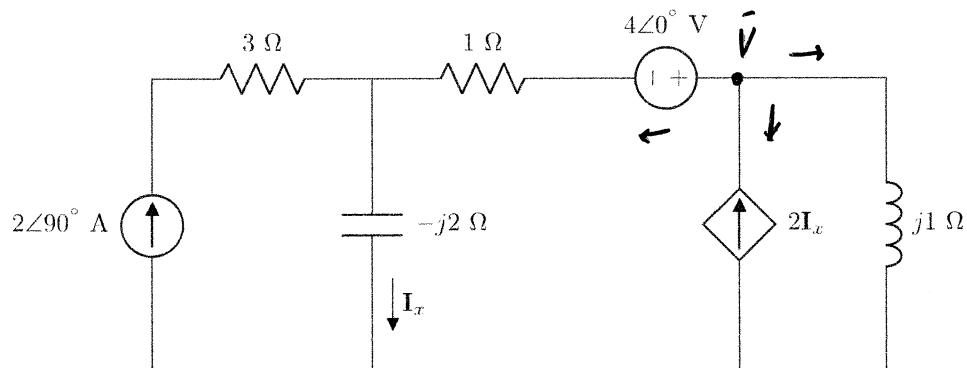
$$\boxed{\bar{V}_1 [-2 - j2] + \bar{V}_2 = 0}$$

$$\bar{c} = -2 - j2$$

$$\bar{d} = 1$$

Question 60 [2]

Consider the circuit diagram below.

60 Find the contribution to I_x from the voltage source in **rectangular** form. [2]

$$\frac{\bar{V} - 4}{1 - j2} - 2\bar{I}_x + \frac{\bar{V}}{j} = 0$$

$$\frac{\bar{V} - 4}{1 - j2} - 2\left(\frac{\bar{V} - 4}{1 - j2}\right) + \frac{\bar{V}}{j} = 0$$

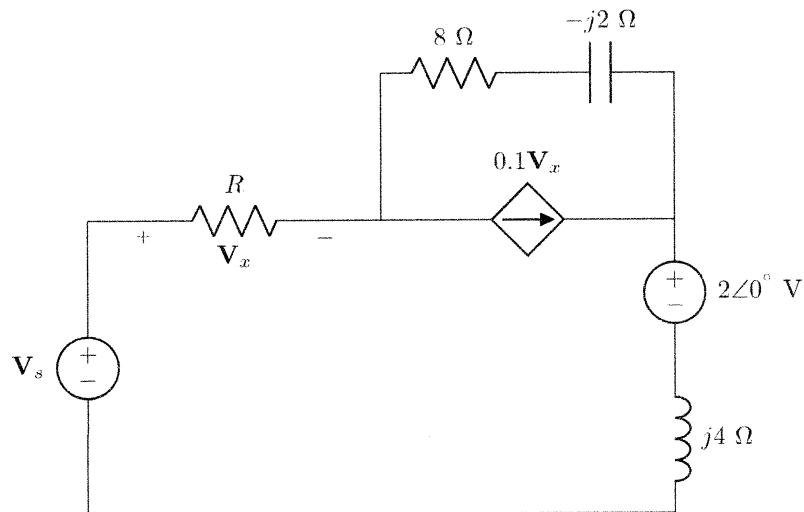
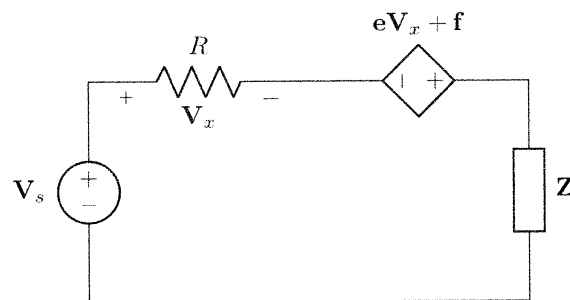
$$\frac{-\bar{V} + 4}{1 - j2} + \frac{\bar{V}}{j} = 0$$

$$\bar{V} \left[\frac{-1}{1 - j2} + \frac{1}{j} \right] = \frac{-4}{1 - j2}$$

$$\bar{V} = 1,2 - j0,4 \text{ V}$$

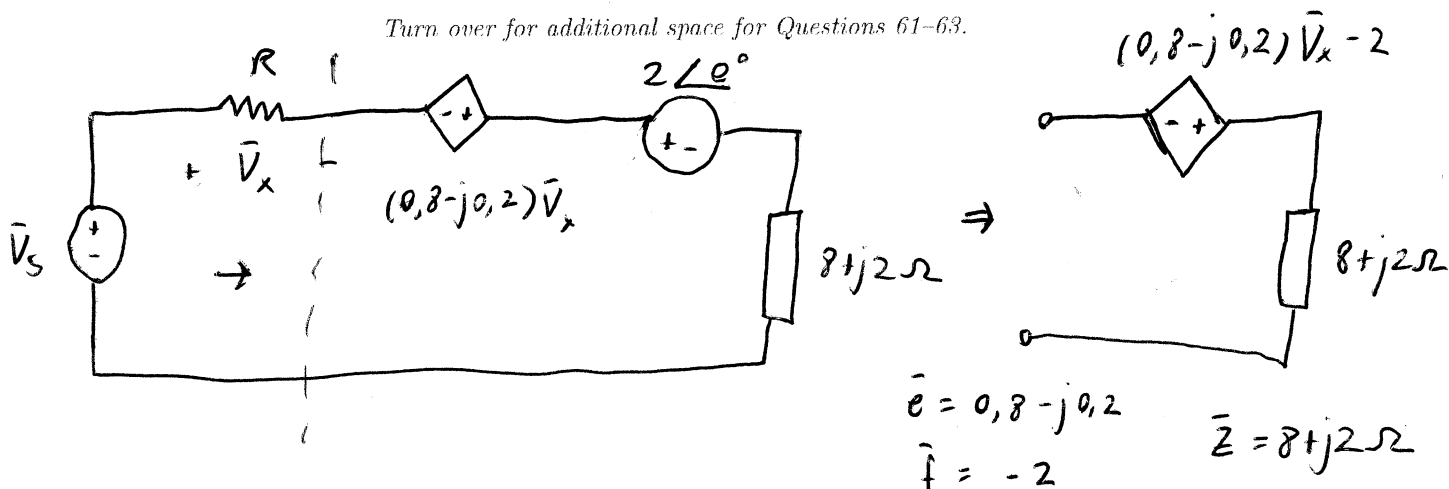
$$\begin{aligned} \bar{I}_x &= \frac{\bar{V} - 4}{1 - j2} \\ &= \frac{1,2 - j0,4 - 4}{1 - j2} \\ &= -0,4 - j1,2 \text{ A} \end{aligned}$$

Questions 61 to 62 [3]

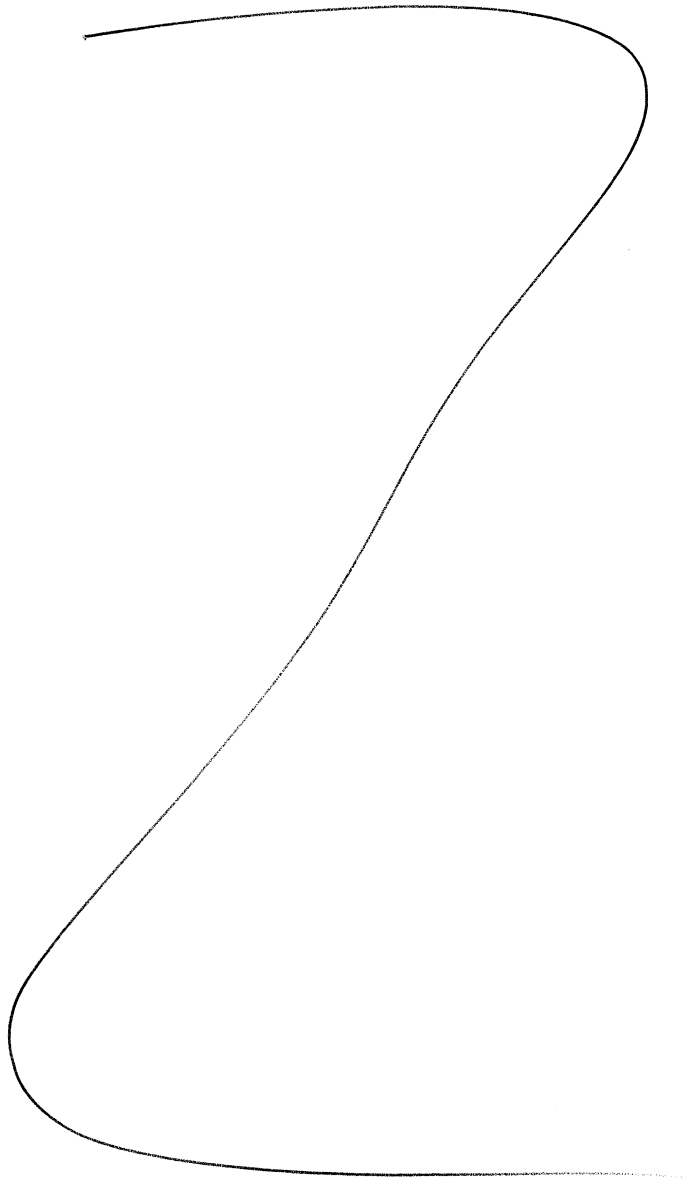
Apply source transformation to transform circuit **A** to circuit **B**.**Circuit A:****Circuit B:**

- | | |
|----|---|
| 61 | Find Z in rectangular form. [1] |
| 62 | Find e in rectangular form. [1] |
| 63 | Find f in rectangular form. [1] |

Turn over for additional space for Questions 61–63.

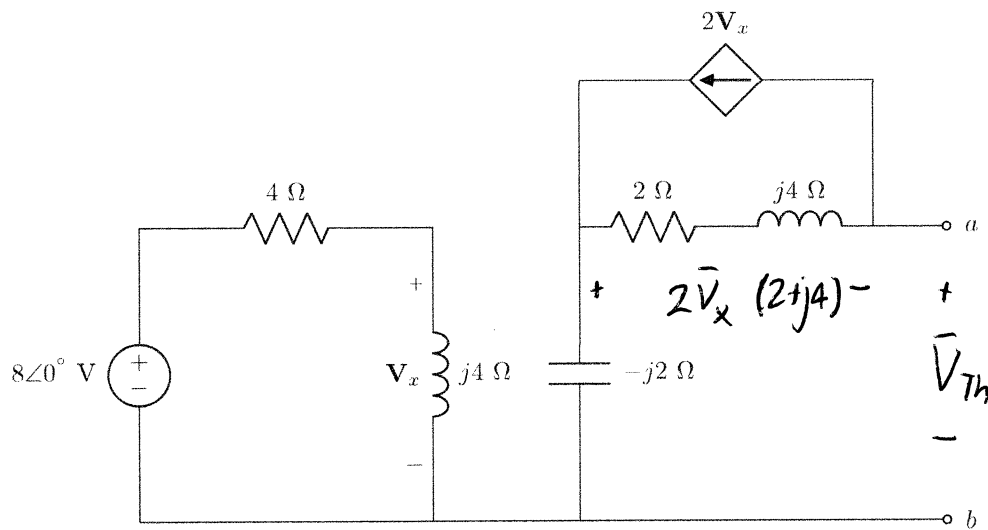


Page for rough work.



Questions 63 to 64 [4]

Consider the circuit diagram below.

Find the Thévenin equivalent voltage V_{Th} and impedance Z_{Th} at terminals $a - b$.63 Determine V_{Th} in rectangular form. [2]64 Determine Z_{Th} in rectangular form. [2]

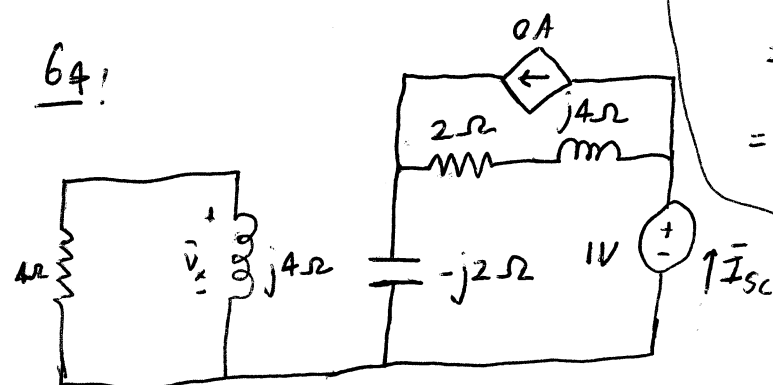
63:
$$\bar{V}_x = 8\angle 0^\circ \left[\frac{j4}{j4 + 4} \right] = 4 + j4 \text{ V}$$

$$\bar{V}_{Th} = -2\bar{V}_x (2 + j4)$$

$$= -2(4 + j4)(2 + j4)$$

$$= 16 - j48 \text{ V}$$

64!



$$\bar{I}_{sc} = \frac{1}{2 + j4 - j2} = 0,25 - j0,25 \text{ A}$$

$$\bar{Z}_{Th} = \frac{1\angle 0^\circ}{\bar{I}_{sc}} = 2 + j2 \text{ } \Omega$$

Total Section 3: [31]

Grand Total: [92]
